Lecture 6: American Derivatives

September 26, 2010

1 Risk-Neutral Evaluation

Under no arbitrage, European options can be effectively evaluated using (conditional) expectations of discounted payoff at maturity. Thus, instead of implementing the backward induction step by step, one can compute the price of an European option using probability distributions.

American options are more complicated than European ones. Holders of an American option can exercise it at any time k, $0 \leq k \leq N$, where N is the maturity. In general, we may assume that the payoff of a given American option at time k, if it is exercised, is G_k, where G_k is $\mathcal{F}_k$-measurable. G_k is called the intrinsic value of the American option. Holders of this option will try to exercise it at an optimal time, so that they can receive the maximum possible discounted payoff from this option. Formally, if we denote by τ a stopping rule for exercise, then at time 0, the expectation of the discounted payoff if we follow this stopping rule is

$$\tilde{E}\left(\mathbb{1}_{\{\tau \leq N\}} \frac{G_\tau}{(1+r)^\tau}\right). \tag{1}$$

The holders want to find a stopping rule τ^* so that

$$\tilde{E}\left(\mathbb{1}_{\{\tau^* \leq N\}} \frac{G_{\tau^*}}{(1+r)^{\tau^*}}\right) = \max_\tau \tilde{E}\left(\mathbb{1}_{\{\tau \leq N\}} \frac{G_\tau}{(1+r)^\tau}\right). \tag{2}$$

The optimal stopping rule τ^* not only maximizes the expectation discounted payoff at time 0, but also does it at any time k. Formally, if τ^* does not stop before time k for some paths, we also have

$$\tilde{E}\left(\mathbb{1}_{\{k \leq \tau^* \leq N\}} \frac{G_{\tau^*}}{(1+r)^{\tau^*-k}} \middle| \mathcal{F}_k\right) = \max_\tau \tilde{E}\left(\mathbb{1}_{\{k \leq \tau \leq N\}} \frac{G_\tau}{(1+r)^{\tau-k}} \middle| \mathcal{F}_k\right). \tag{3}$$

So the optimal exercise stopping rule τ^* is a globally optimal stopping rule. In view of (2) and (3), the price of this American option should be given by the left hand side of (2) or (3), provided that the optimal stopping rule τ^* exists. That is, if τ^* is known, then the time k price of the American option, V_k, is given by

$$V_k = \tilde{E}\left(\mathbb{1}_{\{k \leq \tau^* \leq N\}} \frac{G_{\tau^*}}{(1+r)^{\tau^*-k}} \middle| \mathcal{F}_k\right), \ 0 \leq k \leq N. \tag{4}$$

The formula in (4) is the equivalent of the risk-neutral pricing for European options. We notice from (4) that the price of an American option is still the discounted expectation of future cash flow, if the optimal stopping time has not past.

Recall that we have a backward algorithm for the evaluation of an American option. It is possible to show that the price computed using that method is the same as the one given by (4)[1]. If one can find the optimal stopping rule, then the price V_k can again be easily computed using (4).

[1] The complete proof of this fact is given in Theorem 4.4.2 and Theorem 4.4.3. The basic idea is to show that both the price given by backward induction and the price given by (4) satisfy are smallest possible process that satisfy certain conditions, so they must be equal to each other.

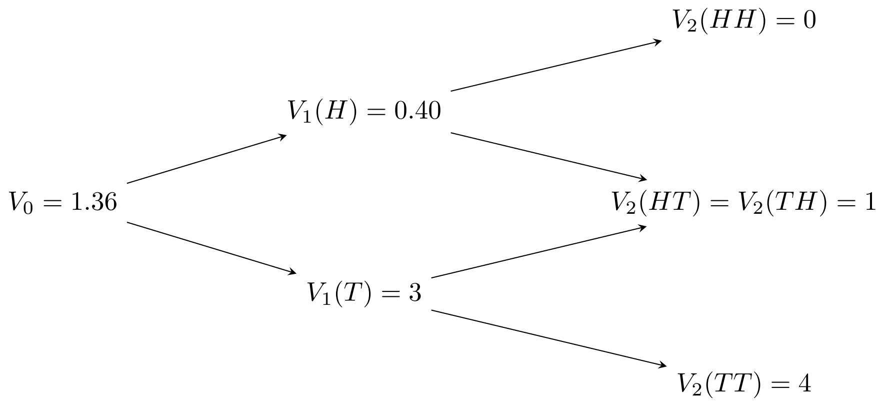

Figure 1: A two-period binomial model.

2 Optimal Exercise Time

To find the optimal exercise time, recall that at any time k, the holder is always comparing the intrinsic value G_k, with the expectation of the discounted payoff at time $k+1$. The maximum of the two gives the price of the American option at time k. In general, the value of an American option is always greater than its intrinsic value. As a holder of an American option, the best thing to do is wait until the intrinsic value is exactly equal to the value of the option. This is optimal because, at such time, the value of the option is exactly equal to the payoff if we exercise it now, and there is no reason to take the risk to keep it any longer.[2]

More specifically, the optimal exercise time τ^* is determined by

$$\tau^* \quad = \quad \min\{k; V_k = G_k\}. \tag{5}$$

To find the optimal stopping rule in a N-period binomial model, we need to know the price process $\{V_k\}$. We cannot use (4) to obtain it since we would suffer circular reasoning. Thanks to the equivalence between backward induction and risk-neutral pricing (4), we can compute the price process $\{V_k\}$ easily with backward algorithm.

Example 1. Recall Example 1 in Lecture 5, the price process $\{V_k\}$ of a call struck at 5 is given in Figure 1. Moreover, it is easily seen that

$$G_0 = (5-4) = 1,\ G_1(H) = (5-8) = -3,\ G_1(T) = (5-2) = 3,$$
$$G_2(HH) = (5-16) = -9,\ G_2(HT) = G_2(TH) = (5-4) = 1,\ G_2(TT) = (5-1) = 4.$$

Using (5) we obtain the optimal stopping rule[3]

$$\begin{aligned} \tau^*(HH) &= \min\phi = \infty, \\ \tau^*(HT) &= 2, \\ \tau^*(TH) &= 1, \\ \tau^*(TT) &= 1. \end{aligned}$$

We knew that for an American option, the discounted price process $\{\frac{V_k}{(1+r)^k}\}$ is not a martingale under the risk-neutral measure. But the stopped process $\{\frac{V_{k\wedge\tau^*}}{(1+r)^{k\wedge\tau^*}}\}$ is a martingale. We have seen this fact in Example 3 of Lecture 5. A rigorous proof can be found in Theorem 4.4.5.

[2] It is true that the actual payoff may be greater if we keep it longer. But it could also be the case that this does not happen. A good stopping rule is not a gambling strategy.

[3] As a convention, minimum of empty set is infinity. This means never exercise the option and let it expire worthless.

3 Evaluation of American Calls

In this section we price an American call option of a stock with no dividends, or equivalently, find the optimal exercise time τ^* for this option. Let us assume the strike price of the call is K and its maturity is N. We will see that, surprisingly, the optimal strategy for a holder of an American call is not to exercise it until the maturity.

Here is a finance (non-math) argument[4]. Notice that, at time N, the payoff of this call plus a bond that pays K at time N is

$$(S_N - K)^+ + K = \max\{S_N, K\} \geq S_N. \tag{6}$$

So at time k, if I setup a portfolio that consists of the above call and the above bond, then I have to spend

$$X_k \quad = \quad V_k + \frac{K}{(1+r)^{N-k}}, \tag{7}$$

where V_k is the time k price of the call option. At time N, the value of this portfolio will dominate the stock price S_N. As a result, no arbitrage implies that at time k,

$$X_k > S_k. \tag{8}$$

Otherwise, I am just going to short one share of stock at time k, and use the proceeds to setup this portfolio; at time N, I have zero probability of losing money, and have a positive probability $\tilde{P}(S_T < K)$ of making money. This is an arbitrage!

Combining (7) and (8) we have that

$$V_k \quad > \quad S_k - \frac{K}{(1+r)^{N-k}} > S_k - K. \tag{9}$$

Since the value of the option is always strictly greater than its intrinsic value $S_k - K$, we see from (5) that the holder should not exercise this option before its maturity N.

We mentioned that American options are usually more expensive than their European counterparts because of early exercise premium. But when it comes to American call, the premium is zero. American calls and European calls have the same price.

Remark 1. *The above proof is very general. The argument holds as long as there is no arbitrage, no dividend payment, and the interest rate is positive.*

Remark 2. *For European calls and puts, we have put-call parity:*

$$C_k - P_k \quad = \quad F_k, \tag{10}$$

where C_k, P_k and F_k are respectively the prices of a call, a put and a forward that have the same strike K and same maturity N. This relationship is a simple consequence of risk-neutral pricing formula. However, we do not have put-call parity for American calls and puts, because the optimal exercise times for calls and puts are usually different.

[4] Check Section 4.5 of the textbook if you want a mathematical argument.